

ON HOLOMORPHIC AND HARMONIC FUNCTION SPACES

E. Chernysh

CONTENTS

1	Holomorphic Spaces $\mathcal{H}(\Omega)$	3
1.1	The Preliminary Case in $\mathbb{R}^d$	3
1.2	Holomorphic Spaces	3
1.3	Compactness in $\mathcal{H}(\Omega)$	6
2	Harmonic Spaces	10
2.1	Characterization of Harmonic Functions	10
2.2	Maximum Principles	12
2.3	Completeness of $\Delta(\Omega)$	13
3	$\mathcal{H}(\Omega)$ as a Banach Space	14
3.1	Convergence on Restrictions of $\mathcal{H}(\Omega)$	14
3.2	The Riemann Mapping Theorem	16

CHAPTER 1

HOLOMORPHIC SPACES $\mathcal{H}(\Omega)$

1.1 THE PRELIMINARY CASE IN $\mathbb{R}^d$

We briefly examine the completeness and convergence properties of sequences of holomorphic functions, with common support $\text{supp}(f) \subseteq \mathbb{C}$. We shall mostly show that the space of all such functions is *complete* and give sufficient conditions for the compactness of a family of such functions $\mathcal{F}$. Ultimately, we give an elegant proof of Montel's theorem and prove the *Riemann Mapping Theorem*.

Given an open set $\Omega \subseteq \mathbb{R}^n$ we shall denote by $\mathcal{D}(\Omega)$ the space of all real valued *differentiable* functions over Ω , given by

$$\mathcal{D}(\Omega) := \{f : \bar{\Omega} \rightarrow \mathbb{R} : f \text{ is differentiable in } \Omega \text{ and continuous on } \partial\Omega\}$$

We shall endow this space with the sup-norm, given by

$$\|\cdot\|_{\infty} := \sup_{x \in \Omega} |\cdot(x)|$$

which is well defined in our case. It is well known that the resulting metric space is not complete. That is, it is not closed under limits. Indeed, consider the sequence defined by

$$f_n(x) := \sqrt{x^2 + \frac{1}{n}}, \quad \Omega := (-1, 1)$$

For each $n \in \mathbb{N}$, this function is continuous on $[-1, 1]$ and differentiable on $(-1, 1)$. Hence, $f_n \in \mathcal{D}((-1, 1))$ for all $n \in \mathbb{N}$. Set $f(x) := |x|$. Note that

$$|x| \leq f_n(x), \quad \forall n \in \mathbb{N} \text{ and } x \in [-1, 1]$$

Moreover, for all $x \in (-1, 1)$ we have necessarily $x^2 \leq |x|$ and thus $x^2 + 1/n \leq |x| + 1/n$ thus implying that $f_n(x) \leq |x| + 1/n$. More precisely,

$$|x| \leq f_n(x) \leq |x| + \frac{1}{n}, \quad \forall n \in \mathbb{N} \text{ and } x \in [-1, 1]$$

It is clear from this above estimate that $f_n \rightarrow f$ uniformly with respect to $\|\cdot\|_{\infty}$. However, f is not differentiable at 0, proving that $\mathcal{D}(\Omega)$ is not necessarily complete.

1.2 HOLOMORPHIC SPACES

We now turn towards holomorphic counterparts. Let us first recall Cauchy's theorem:

THEOREM 1. *Let $\Omega \subseteq \mathbb{C}$ be an open set and $\gamma \subset \Omega$ a simple closed curve contained along with its interior inside Ω . Then, if $f : \Omega \rightarrow \mathbb{C}$ is holomorphic in Ω*

$$\oint_{\gamma} f(\zeta) d\zeta = 0 \tag{1.1}$$

THEOREM 2 (Morera). *Let $\Omega \subseteq \mathbb{C}$ be a connected open set and $f : \Omega \rightarrow \mathbb{C}$ be continuous in Ω . Suppose further that for all simple closed curves Γ contained in Ω along with their interiors*

$$\oint_{\Gamma} f(\zeta) d\zeta = 0$$

Then f is holomorphic in Ω .

Proof. We shall construct a function $F : \Omega \rightarrow \mathbb{C}$ that is holomorphic and satisfies $F' = f$ for all $z \in \Omega$. To do this, fix a point $z_0 \in \Omega$ and consider any $z \in \Omega$. Given any simple curve γ beginning at z_0 and ending at z , we define a function

$$F(z) := \int_{\gamma} f(\zeta) d\zeta$$

We claim that this is well defined. Certainly, if γ_1 and γ_2 are two such curves, since they share end points, if we let $\Gamma := \gamma_1 \cup \gamma_2^-$, where γ_2^- is the same curve with reverse orientation, then Γ is a simple closed curve in Ω . Hence, by hypothesis

$$\oint_{\Gamma} f(\zeta) d\zeta = \int_{\gamma_1} f(\zeta) d\zeta + \int_{\gamma_2^-} f(\zeta) d\zeta = 0$$

Now, we show that F is holomorphic in Ω with $F' \equiv f$ in Ω . Indeed, let $h \in \mathbb{C}$ with $|h| > 0$. Now take h so small that $z + h \in \Omega$. Then,

$$F(z + h) - F(z) = \int_z^{z+h} f(\zeta) d\zeta$$

where this integral is evaluated along any curve contained in Ω connecting z and $z + h$. Since this domain is connected for all h small enough we may take a straight line connecting z and $z + h$ contained entirely in Ω . By continuity of f , for ζ near z we may also write

$$f(\zeta) = f(z) + \varphi(\zeta), \quad \varphi(\zeta) \xrightarrow{\zeta \rightarrow z} 0$$

Thence,

$$F(z + h) - F(z) = \int_z^{z+h} (f(z) + \varphi(\zeta)) d\zeta \leq f(z)h + \int_z^{z+h} \varphi(\zeta) d\zeta$$

Whereby

$$\left| \frac{F(z_0 + h) - F(z_0)}{h} - f(z_0) \right| \leq \frac{1}{|h|} \left| \int_z^{z+h} \varphi(\zeta) d\zeta \right| \leq \sup_{\zeta} |\varphi(\zeta)|$$

where the sup is taken over ζ with $|\zeta - z| < h$. Hence, letting $h \rightarrow 0$ we conclude that $F'(z) = f(z)$, that F is holomorphic in Ω and $F'(z) = f(z)$ in Ω . Since F'' exists, we have that f is holomorphic. ■

One can view Morera's theorem as a converse to Cauchy's theorem. We may now prove our main topic. First, given an open set $\Omega \subseteq \mathbb{C}$ we denote by $\mathcal{H}(\Omega)$ the collection of all functions $f : \Omega \rightarrow \mathbb{C}$ holomorphic on Ω . Unlike the case for $\mathcal{D}(\Sigma)$ for $\Sigma \subseteq \mathbb{R}^n$, this space is complete when endowed with the sup-norm; which unlike the case for real functions, we define

$$\|\cdot\|_{\infty} := \sup_{\xi \in \Omega} |\cdot(\xi)| \tag{1.2}$$

Therefore we have the following definition:

DEFINITION 1. Let $\Omega \subseteq \mathbb{C}$ be open, denote by $H(\Omega)$ the collection of all holomorphic functions $f : \Omega \rightarrow \mathbb{C}$. We shall denote by $\mathcal{H}(\Omega)$ the space $H(\Omega)$ endowed with usual scalar multiplication over $\mathbb{C}$, addition and the sup-norm $\|\cdot\|_\infty$.

Sometimes, given a sequence (f_n) and a point f in $\mathcal{H}(\Omega)$ we shall say f_n converges to f in the space to emphasize that the convergence occurs with respect to the norm $\|\cdot\|_\infty$. We may also write

$$f_n \xrightarrow{\mathcal{H}(\Omega)} f, \quad \text{as } n \rightarrow \infty$$

THEOREM 3. Let $\Omega \subseteq \mathbb{C}$ be connected and open. The space $\mathcal{H}(\Omega)$ is complete when equipped with $\|\cdot\|_\infty$.

Proof. Let $(f_n)_{n \in \mathbb{N}} \subset \mathcal{H}(\Omega)$ be a Cauchy sequence. For each $\varepsilon > 0$ there exists N so that whenever $n, m \geq N$ one has

$$|f_n(z) - f_m(z)| \leq \|f_n - f_m\|_\infty < \varepsilon, \quad \forall n, m \geq N$$

hence $(f_n(z))_{n \in \mathbb{N}}$ is Cauchy in $\mathbb{C}$ for all $z \in \Omega$. Since the complex numbers are complete, there exists a function $f : \Omega \rightarrow \mathbb{C}$ so that

$$\lim_{n \rightarrow \infty} f_n(z) = f(z) \quad \forall z \in \Omega$$

Moreover, since the sequence $(f_n)_{n \in \mathbb{N}}$ is Cauchy in our space for all n, m large we have

$$|f_n(z) - f_m(z)| \leq \|f_n - f_m\|_\infty < \varepsilon$$

passing to the limit in $m \rightarrow \infty$ we obtain

$$|f_n(z) - f(z)| \leq \varepsilon$$

implying $\|f_n - f\|_\infty \leq \varepsilon$, i.e. that $f_n \rightarrow f$ uniformly on Ω . As the uniform limit of a continuous function, we conclude that f is continuous over Ω . It remains only to show that f is holomorphic in Ω . Let Γ be a closed curve contained entirely in Ω , along with its interior. Note that by *Cauchy's Theorem* we have

$$\oint_{\Gamma} f_n(\zeta) d\zeta = 0 \quad \forall n \in \mathbb{N} \quad (\star)$$

Hence,

$$\lim_{n \rightarrow \infty} \oint_{\Gamma} f_n(\zeta) d\zeta = 0 \quad (1.3)$$

Now, let $\varepsilon > 0$ be given. There exists $N \in \mathbb{N}$ so that for all $n \geq N$ one has

$$\|f_n - f\| < \frac{\varepsilon}{\ell(\Gamma)}$$

where $\ell(\Gamma) > 0$ denotes the length of the curve Γ . Then,

$$\begin{aligned} \left| \oint_{\Gamma} f_n(\zeta) d\zeta - \oint_{\Gamma} f(\zeta) d\zeta \right| &\leq \oint_{\Gamma} |f_n(\zeta) - f(\zeta)| d\zeta \\ &\leq \oint_{\Gamma} \|f_n - f\|_\infty d\zeta \\ &\leq \varepsilon \end{aligned}$$

Hence,

$$\lim_{n \rightarrow \infty} \oint_{\Gamma} f_n(\zeta) d\zeta = \oint_{\Gamma} f(\zeta) d\zeta$$

Implying that $\oint_{\Gamma} f(\zeta) d\zeta = 0$. Since the curve $\Gamma \subset \Omega$ was arbitrary, we conclude by *Morera's theorem* that $f \in \mathcal{H}(\Omega)$. ■

1.3 COMPACTNESS IN $\mathcal{H}(\Omega)$

We adopt the usual topological definition of compactness. Given an appropriate domain $\Omega \subseteq \mathbb{C}$, we say a subset $\mathcal{F} \subseteq \mathcal{H}(\Omega)$ is **compact** if for any open cover $\{\mathcal{G}_j\}_{j \in \mathbb{N}}$ of $\mathcal{F}$ there exists a finite sub-cover $\{\mathcal{G}_j\}_{j=1}^N$ of $\mathcal{F}$.

We shall now refer to sets $\Omega \subset \mathbb{C}$ as **proper** if $\mathbb{C} \setminus \Omega \neq \emptyset$. It is evident that if subspace $\mathcal{F} \subset \mathcal{H}(\Omega)$ is compact, then it must consist of a family of functions closed under limits and are uniformly bounded, with respect to the norm $\|\cdot\|_\infty$. That is, there exists a constant $M > 0$ so that $\|f\|_\infty \leq M$ for all $f \in \mathcal{F}$.

Ultimately, we establish *Montel's Theorem*. We require the following definition

DEFINITION 2 (Equicontinuity). *Let $\mathcal{F}$ consist of a family of functions defined on a set $\Lambda \subseteq \mathbb{C}$. We shall call the family $\mathcal{F}$ **equicontinuous** if for all $\varepsilon > 0$ there exists $\delta > 0$ so that*

$$|f(x) - f(y)| < \varepsilon \quad (1.4)$$

for all $f \in \mathcal{F}$ whenever $|x - y| < \delta$ for $(x, y) \in \Lambda \times \Lambda$. In symbolic terms:

$$(\forall \varepsilon > 0) (\exists \delta > 0) (\forall x, y \in \Lambda \wedge \forall f \in \mathcal{F}) (|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon) \quad (\mathcal{E})$$

Note that any function that is equicontinuous is also *uniformly continuous*. An example is in order. Suppose $\mathcal{F}$ consists of the set of all functions defined on $[0, 1]$, continuous, and differentiable on an open set containing $[0, 1]$ with bounded first derivative. This is an equicontinuous family. Indeed, let M be a uniform bound on the first derivatives of the elements of $\mathcal{F}$. Fix $\varepsilon > 0$, and let $\delta := \varepsilon/M > 0$. Then, by the mean value theorem given any pair $x, y \in [0, 1]$ with $x \neq y$ there exists a point c between x and y so that

$$|f'(c)| = \frac{|f(x) - f(y)|}{|x - y|}, \forall f \in \mathcal{F}$$

Thence, using the bound on f' that is independent of f , we have

$$|f(x) - f(y)| \leq M |x - y|$$

which proves the desired conclusion whenever $|x - y| < \delta$ and holds for all $f \in \mathcal{F}$.

Now to give the crucial definition of this section:

DEFINITION 3. *Given appropriate $\Omega \subseteq \mathbb{C}$ a subspace $\mathcal{F} \subseteq \mathcal{H}(\Omega)$ is called **normal** if every sequence $(f_n)_n$ in $\mathcal{F}$ has a subsequence that converges uniformly on every compact subset of Ω .*

The reader should note that $\mathcal{F}$ is normal if and only if it is pre-compact in $\mathcal{H}(\Omega)$. We have the following theorem:

THEOREM 4 (Montel). *Let $\mathcal{F}$ be a subspace of $\mathcal{H}(\Omega)$ and is uniformly bounded on every compact subset Λ of Ω . Then,*

(i) $\mathcal{F}$ is equicontinuous on each Λ .

(ii) $\mathcal{F}$ is normal.

The proof is quite long, and we shall deconstruct it elegantly into lemmata. First, we would like to show that this is special to holomorphic functions, and does not necessarily apply to real differentiable functions. Certainly, consider the family of functions on $(-1, 1)$ defined by $f_n(x) := \sin nx$. These are all differentiable (even infinitely differentiable) on this domain and uniformly bounded by 1. However, this family is not equicontinuous. To see this we argue by contradiction. Suppose the family is equicontinuous and consider the point $x = 0$. For $\varepsilon := 1/2 > 0$ there must be $\delta > 0$ so that for all f_n one has $|f(x) - f(y)| < 1/2$ whenever

$|x - y| < \delta$ for $x, y \in (-1, 1)$. In particular, for any $0 < x < \delta$ one has $|\sin nx| < 1/2$. However, if we let $x = \frac{\pi}{2N}$ for N so that $x < \delta$ then we are left with

$$|f_N(x)| = \left| \sin \frac{\pi}{2} \right| = 1 > 1/2$$

which is a contradiction. It also has no uniformly convergent subsequence on all compact subsets of $(-1, 1)$ by a similar argument.

DEFINITION 4. Let $\Omega \subseteq \mathbb{C}$. A sequence of compact sets $\{\Lambda_n\}_{n \in \mathbb{N}}$ with $\Lambda_n \subseteq \Lambda_{n+1}$ for all n is said to be an **exhaustion** of Ω provided $\Omega = \bigcup_{n \in \mathbb{N}} \Lambda_n$.

LEMMA 5. Let $\Omega \subseteq \mathbb{C}$ be an open set. This set has an exhaustion.

Proof. First assume that Ω is bounded. Then, for $n \in \mathbb{N}$ define

$$\Lambda_n := \left\{ \xi \in \Omega : d(\xi, \partial\Omega) \geq \frac{1}{n} \right\}$$

where it is clear that $\Omega = \bigcup_{n \in \mathbb{N}} \Lambda_n$; else we need only replace the above with

$$\Lambda_n := \left\{ \xi \in \Omega : d(\xi, \partial\Omega) \geq \frac{1}{n}, |\xi| \leq n \right\}$$

■

LEMMA 6. Let $\Omega \subset \mathbb{C}$ be a proper open set and let $\Lambda \subset \Omega$ be compact. Then, there exists $\rho > 0$ so that $B(z, \rho) \subseteq \Omega$ for all $z \in \Lambda$.

Proof. By way of contradiction, suppose that this is false. Then, for each $n \in \mathbb{N}$ we may select a point $z_n \in \Lambda$ so that for all $B(z_n, 1/n) \cap \Omega^c \neq \emptyset$. Then, in this way we construct two sequences $(z_n), (\zeta_n)$ with $z_n \in \Lambda$ and $\zeta_n \in \Omega^c$ for all n satisfying

$$|z_n - \zeta_n| < \frac{1}{n}$$

By compactness of Λ , we may extract a subsequence (z_{n_k}) converging to $z \in \Lambda$ so that the above still holds true for all indices $n_k \in \mathbb{N}$ indexed by k . To see that $\zeta_{n_k} \rightarrow z$ as well in k we need only note that for all k :

$$|\zeta_{n_k} - z| \leq |\zeta_{n_k} - z_{n_k}| + |z_{n_k} - z| \leq \frac{1}{n_k} + |z_{n_k} - z|$$

where letting $k \rightarrow \infty$ shows that $\lim_{k \rightarrow \infty} \zeta_{n_k} = z$. But, Ω^c is closed implying that $z \in \Omega^c$ as well, i.e. $\Omega \cap \Omega^c \neq \emptyset$: a clear contradiction.

■

LEMMA 7. Let $\Lambda \subset \mathbb{C}$ be compact and $(w_j)_{j \in \mathbb{N}}$ a countable, dense subset of $\mathbb{C}$. For any $\delta > 0$ there exist finitely many $w_1, \dots, w_N$ so that $\Lambda \subseteq \bigcup_{j=1}^N B(w_j, \delta)$ where $B(w_j, \delta)$ denotes the open ball of radius δ about w_j .

Proof. Otherwise, for some $\delta > 0$ one would have $\Lambda \not\subseteq \bigcup_{j=1}^N B(w_j, \delta)$ for all $N \in \mathbb{N}$. Now, there must exist $z_1 \in \Lambda$ such that $z_1 \notin B(w_1, \delta)$. Similarly, there must be $z_2 \in \Lambda$ with $z_1, z_2 \notin B(w_1, \delta) \cup B(w_2, \delta)$ and so on. That is, there exists a sequence $(z_n)_{n=1}^\infty$ of elements in Λ satisfying

$$z_m \notin \bigcup_{j=1}^n B(w_j, \delta), \quad \forall n \geq m \tag{1.5}$$

By compactness of Λ , we may extract a subsequence $(z_k)_{k=1}^\infty$ converging to some $z \in \Lambda$. Since $(w_j)_{j \in \mathbb{N}}$ is dense in $\mathbb{C}$, we may select a sequence $(w_\ell)_{\ell \in \mathbb{N}}$ converging to z . More precisely, we may find $N_1, N_2 \in \mathbb{N}$ such that

$$|z_k - z| < \frac{\delta}{2}, \quad \forall k \geq N_1 \quad (1.6)$$

$$|w_\ell - z| < \frac{\delta}{2}, \quad \forall \ell \geq N_2 \quad (1.7)$$

Now let $N := \max(N_1, N_2)$ and suppose $k, \ell \geq N$ with $k \leq \ell$. Then,

$$|z_k - w_\ell| \leq |z_k - z| + |z - w_\ell| < \delta$$

i.e. $z_k \in B(w_\ell, \delta)$: a contradiction. ■

Proof of Montel's Theorem. We can now establish the result from (i). Fix a compact subset Λ of Ω , $f \in \mathcal{F}$, and let $\rho > 0$ be such that $B(z, 3\rho) \subseteq \Omega$ for all $z \in \Lambda$, which we know to exist by the previous lemma. Now let $z, w \in \Lambda$ with $|z - w| < \delta := \rho$ and let Γ denote the circumference of the circle of radius $B(w, 2\rho) \subseteq \Omega$. Observe that z also lies in the interior of this circle. Since f is holomorphic in Ω we may express:

$$f(z) - f(w) = \frac{1}{2\pi i} \oint_{\Gamma} f(\zeta) \left(\frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right) d\zeta$$

So that

$$|f(z) - f(w)| \leq \frac{1}{2\pi} \oint_{\Gamma} |f(\zeta)| \left| \frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right| |d\zeta|$$

Note that $|\zeta - z| \geq |\zeta - w| - |z - w| \geq \rho$. We now calculate

$$\left| \frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right| = \frac{|z - w|}{|\zeta - z| |\zeta - w|} = \frac{|z - w|}{2\rho |\zeta - z|} \leq \frac{|z - w|}{\rho^2}$$

Therefore, if we denote by $M > 0$ the uniform bound on the family $\mathcal{F}$

$$|f(z) - f(w)| \leq \frac{M}{2\pi} \cdot \frac{4\pi\rho}{\rho^2} \cdot |z - w| = C |z - w|, \quad \forall f \in \mathcal{F}$$

for some positive constant C independent of $f \in \mathcal{F}$. This establishes that $\mathcal{F}$ is equicontinuous on any compact subset Λ of Ω as was claimed in (i).

We shall now show (ii). First observe that the *complex rationals* $\mathbb{Q}_{\mathbb{C}} := \{x + iy : x, y \in \mathbb{Q}\}$ forms a countable, dense subset of $\mathbb{C}$. Indeed, it suffices to approximate any complex $z = x + iy$ for real x, y by elements of $\mathbb{Q}_{\mathbb{C}}$. There exist rational sequences $(x_n), (y_n)$ such that $(x_n) \rightarrow x$ and $(y_n) \rightarrow y$. Clearly, $x_n + iy_n \in \mathbb{Q}_{\mathbb{C}}$ for all n and thus we have found a sequence (z_n) of elements in $\mathbb{Q}_{\mathbb{C}}$ converging to $z \in \mathbb{C}$.

Now let $(f_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{F}$, fix a compact subset Λ of the global set Ω and enumerate the complex rationals by $(w_j)_{j \in \mathbb{N}}$. Since the sequence $(f_n)_{n \in \mathbb{N}}$ is uniformly bounded by a constant $M > 0$, we may select a subsequence $(f_{n,1})_{n \in \mathbb{N}}$ so that $f_{n,1}(w_1)$ converges to a complex number as $n \rightarrow \infty$.

From this subsequence, we may also recover a subsequence $(f_{n,2})_{n \in \mathbb{N}}$ so that $f(w_2)$ converges in $\mathbb{C}$; and as a subsequence of a convergent subsequence one easily sees $f_{n,2}(w_1)$ converges to the same limit as before. In this way we proceed, inductively constructing subsequences $f_{n,j}$ so that $f_{n,j}(w_k)$ converges for $1 \leq k \leq j$ and $j \in \mathbb{N}$ as $n \rightarrow \infty$.

Now define $(g_n)_{n \in \mathbb{N}}$ where $g_n := f_{n,n}$. Clearly, $g_n(w_j)$ converges for all $j \in \mathbb{N}$ as $n \rightarrow \infty$ by construction. Let $\varepsilon > 0$ be given, since the subspace $\mathcal{F}$ is equicontinuous, there exists positive δ such that for any pair $x, y \in K$:

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon, \quad \forall f \in \mathcal{F}$$

Since Λ is compact, it is totally bounded and we may find finitely many $w_1, \dots, w_K$ in our dense collection so that $\Lambda \subseteq B(w_1, \delta) \cup \dots \cup B(w_K, \delta)$. Since each individual $g_n(w_j)$ is a convergent sequence of complex numbers, it is Cauchy, and there exists N so large that for all $n, m \geq N$:

$$|g_n(w_j) - g_m(w_j)| < \varepsilon, \quad \forall 1 \leq j \leq K$$

by taking the maximum of finitely many such N . Now let $z \in \Lambda$ be given. It follows that $z \in B(w_i, \delta)$ for some index $i = 1, \dots, K$. Therefore, if $n, m \geq N$:

$$\begin{aligned} |g_n(z) - g_m(z)| &\leq |g_n(z) - g_n(w_i)| + |g_n(w_i) - g_m(z)| \\ &\leq |g_n(z) - g_n(w_i)| + |g_n(w_i) - g_m(w_i)| + |g_m(w_i) - g_m(z)| \\ &\leq \varepsilon + \varepsilon + \varepsilon = 3\varepsilon \end{aligned}$$

Now, we see that for all $z \in \Lambda$ the sequence $(g_n(z))_{n \in \mathbb{N}}$ is Cauchy, and hence there exists a function $g : \Lambda \rightarrow \mathbb{C}$ so that $g_n(z) \rightarrow g(z)$ point-wise on Λ . To see that the convergence here is uniform, note that by passing to the limit in $m \rightarrow \infty$ above we recover

$$\|g_n - g\|_\infty \leq 3\varepsilon, \quad \forall n \geq N$$

on the compact subset Λ of Ω .

We must now extend this to any compact subset of Ω . By our first lemma, we may select an exhaustion $\{\Lambda_\ell\}_{\ell \in \mathbb{N}}$ of Ω . For each $\ell \in \mathbb{N}$, we repeat the above process and extract a subsequence $g_{n,\ell}$ converging uniformly on Λ_ℓ . Then, the final subsequence $(h_n)_{n \in \mathbb{N}}$ where $h_n := g_{n,n}$ converges uniformly on any such Λ_ℓ for $\ell \in \mathbb{N}$.

Now, if K is any compact subset of Ω it must be contained in some Λ_α for $\alpha \in \mathbb{N}$, for otherwise we would have $K \not\subseteq \Lambda_\ell$ for all $\ell \in \mathbb{N}$, whence $K \not\subseteq \bigcup_{\ell \in \mathbb{N}} \Lambda_\ell = \Omega$; a contradiction. The theorem has been proven and we are done. ■

We now extend our definition. Given a proper, open and connected set $\Omega \subset \mathbb{C}$ and a compact set $\Lambda \subseteq \mathbb{C}$ we define $\mathcal{H}(\Omega | \Lambda)$ to be the functions in $\mathcal{H}(\Omega)$ viewed as functions over the compact subset Λ .

We may now give the following result

THEOREM 8. *Let $\Lambda \subset \Omega \subseteq \mathbb{C}$ be compact. Suppose $\mathcal{F} \subseteq \mathcal{H}(\Omega | \Lambda)$. The following are equivalent:*

- (i) $\mathcal{F}$ is a compact subspace.
- (ii) $\mathcal{F}$ is closed and bounded.

Proof. (i $\implies$ ii). Let $\mathcal{F}$ be a compact subspace. Then, it is closed and bounded by traditional metric space theory.

(ii $\implies$ i). Suppose that $\mathcal{F}$ is closed and bounded and choose a sequence $(f_n)_{n \in \mathbb{N}}$ in $\mathcal{F}$. This is also a sequence in $\mathcal{H}(\Omega)$. By the argument in Theorem (4) (i) it follows that $\mathcal{F}$ is equicontinuous on $\Lambda \subset \Omega$ and hence an application of (ii) yields a subsequence that converges uniformly on Λ . Since $\mathcal{F}$ is closed, this limit must also live in $\mathcal{F}$. Ergo, $\mathcal{F}$ is compact in the topological sense. ■

CHAPTER 2

HARMONIC SPACES

Let $\Omega \subseteq \mathbb{R}^d$ be an open set, we first define an operator on the space $\mathcal{C}^2(\Omega)$

$$\Delta := \sum_{j=1}^d \frac{\partial^2}{\partial x_j^2} \quad (2.1)$$

This is called the **Laplacian Operator**. Given such a set, we define $\Delta(\Omega)$ to be the space of all functions $u \in C^2(\Omega)$ such that $\Delta u \equiv 0$ in Ω , which we endow with the usual sup-norm:

$$\|\cdot\|_\infty := \sup_{x \in \Omega} |\cdot(x)|$$

2.1 CHARACTERIZATION OF HARMONIC FUNCTIONS

Here we discuss the more interesting results involving harmonic functions. Recall that a couple dozen lines before, we mentioned the averaging properties. It turns out that harmonic functions behave quite nicely. We will see that if a function u is harmonic in a domain Ω (and of class $C^2(\Omega) \cap C(\overline{\Omega})$), then for any ball $B(x, \rho) \subset \Omega$ one has both

$$\oint_{B(x, \rho)} u(y) dy = u(x) = \oint_{\partial B(x, \rho)} u(y) dS \quad (2.2)$$

Here the $\oint$ symbol denotes the *mean integral*. That is, we take the integral and divide by the volume of the ball $B(x, \rho)$ (or $\partial B(x, \rho)$ in the other case):

$$\oint_{B(x, \rho)} u(y) dy = \frac{1}{\text{Vol}(B(x, \rho))} \int_{B(x, \rho)} u(y) dy \quad (2.3)$$

In general, for a domain $\Sigma \subseteq \mathbb{R}^n$ and an integrable function $f : \Sigma \rightarrow \mathbb{R}$ we define

$$\oint_\Sigma f := \frac{1}{\mu(\Sigma)} \int_\Sigma f$$

Where μ denotes the *Lebesgue measure* in $\mathbb{R}^n$.

The converse is also true, if equation (2.2) is satisfied in a domain Ω for **any** ball then the function is also harmonic in this domain. We state this below as a theorem:

THEOREM 9 (Mean Value Theorem). *Let $\Omega \subset \mathbb{R}^n$ be a domain and let $u \in C^2(\Omega) \cap C(\overline{\Omega})$. The following are equivalent:*

1. u is harmonic in Ω (i.e $\Delta u \equiv 0$ in Ω)
2. For any ball $B(x, \rho) \subset \Omega$:

$$\oint_{B(x, \rho)} u(y) dy = u(x) = \oint_{\partial B(x, \rho)} u(y) dS$$

Proof. (1 $\implies$ 2). Let u be harmonic and pick an arbitrary ball $B(x, \rho) \subset \Omega$ (hence $x \in \Omega$). We will first consider the integral over $\partial B(x, \rho)$. Recall now the *Divergence Theorem* which we will require:

$$\int_S \Delta f(y) dy = \int_{\partial S} \nabla f(y) \cdot d\mathbf{S} \quad (\delta)$$

where $d\mathbf{S} = \nu dS$ and ν is the outwards normal of the ball at a point. Then, since $\Delta u \equiv 0$ in Ω and hence in $B(x, \rho)$:

$$\begin{aligned} 0 &= \int_{B(x, \rho)} \Delta u(y) dy = \int_{\partial B(x, \rho)} \nabla u(y) \cdot d\mathbf{S} = \int_{\partial B(x, \rho)} \nabla u(y) \cdot \nu dS \\ &= \int_{\partial B(0, 1)} \nabla u(x + \rho z) \cdot \nu \rho^{n-1} dS = \rho^{n-1} \int_{\partial B(0, 1)} \nabla u(x + \rho z) \cdot \nu dS(z) \\ &= \rho^{n-1} \int_{\partial B(0, 1)} \nabla u(x + \rho z) \cdot z dS(z) = \rho^{n-1} \int_{\partial B(0, 1)} \frac{\partial}{\partial \rho} [u(x + \rho z)] dS(z) \\ &= \rho^{n-1} \frac{\partial}{\partial \rho} \int_{\partial B(0, 1)} u(x + \rho z) dS(z) = \rho^{n-1} \frac{\partial}{\partial \rho} \left[\frac{1}{\omega_{n-1} \rho^{n-1}} \int_{\partial B(0, 1)} u(x + \rho z) \rho^{n-1} dS(z) \right] \\ &= \rho^{n-1} \frac{\partial}{\partial \rho} \left[\frac{1}{\omega_{n-1} \rho^{n-1}} \int_{\partial B(x, \rho)} u(y) dS \right] \end{aligned}$$

Since $\rho^{n-1} \neq 0$ everywhere we conclude that

$$\frac{\partial}{\partial \rho} \oint_{\partial B(x, \rho)} u(y) dS = \frac{\partial}{\partial \rho} \left[\frac{1}{\omega_{n-1} \rho^{n-1}} \int_{\partial B(x, \rho)} u(y) dS \right] = 0 \quad (2.4)$$

for all $\rho > 0$ and is hence a constant function of ρ . Taking the limit as $\rho \rightarrow 0$ implies

$$\oint_{\partial B(x, \rho)} u(y) dS = \lim_{\rho \rightarrow 0^+} \oint_{\partial B(x, \rho)} u(y) dS = u(x)$$

For the remainder of (2) write

$$\begin{aligned} \oint_{B(x, \rho)} u(y) dy &= \frac{1}{\omega_n \rho^n} \int_0^\rho \left(\int_{\partial B(x, \varepsilon)} u(y) dS(y) \right) d\varepsilon = \frac{1}{\omega_n \rho^n} \int_0^\rho n \omega_n u(x) \varepsilon^{n-1} d\varepsilon \\ &= \frac{u(x)}{\rho^n} \int_0^\rho n \varepsilon^{n-1} d\varepsilon \\ &= u(x) \end{aligned}$$

(2 $\implies$ 1). Suppose that (2) holds but that (1) is false: $\Delta u \not\equiv 0$ in Ω . So there exists a point $x_0 \in \Omega$ so that $\Delta u(x_0) \neq 0$. We may assume without loss of generality that $\Delta u(x_0) = \delta > 0$. Then, we may take a small ball $B(x_0, \rho) \subset \Omega$ over which $\Delta u > 0$ since u is of class C^2 and consequently we have that Δu is continuous inside Ω . Then, we see by our assumption in (2) that for all sufficiently small ρ (i.e whenever $B(x_0, \rho) \subset \Omega$):

$$\frac{\partial}{\partial \rho} \oint_{\partial B(x_0, \rho)} u(y) dS = 0$$

On the other hand,

$$\begin{aligned} \oint_{\partial B(x_0, \rho)} u(y) dS(y) &= \frac{1}{n \omega_{n-1} \rho^{n-1}} \int_{\partial B(x_0, \rho)} u(y) dS(y) \\ &= \frac{1}{n \omega_{n-1} \rho^{n-1}} \int_{\partial B(0, 1)} u(x_0 + \rho z) \rho^{n-1} dS(z) \\ &= \frac{1}{n \omega_{n-1}} \int_{\partial B(0, 1)} u(x_0 + \rho z) dS(z) \end{aligned}$$

Whence,

$$\begin{aligned} \frac{\partial}{\partial \rho} \oint_{\partial B(x_0, \rho)} u(y) dS(y) &= \frac{1}{n\omega_{n-1}} \int_{\partial B(0,1)} \nabla u(x_0 + \rho z) \cdot z dS(z) \\ &= \frac{1}{n\omega_{n-1}} \int_{\partial B(0,1)} \nabla u(x_0 + \rho z) \cdot \nu dS(z) \\ &= \oint_{\partial B(x_0, \rho)} \nabla u(y) \cdot \nu dS(y) \end{aligned}$$

Hence, by multiplying both sides by appropriate constants we deduce that

$$\oint_{B(x_0, \rho)} \Delta u(y) dS(y) = 0, \quad \forall \rho \text{ small} \quad (2.5)$$

This contradicts that $\Delta u > 0$ in a small neighbourhood of x_0 . The theorem is then proven. ■

2.2 MAXIMUM PRINCIPLES

With the help of these theorems we may now deduce the so called *Strong Maximum Principle*. This is a very powerful theorem, which is greatly used in classical and complex analysis (remember all that the real and imaginary parts of holomorphic functions are harmonic!). The idea is that if a function u is harmonic in a domain Ω , it can only achieve its maximum on the boundary, unless it is constant.

THEOREM 10 (Strong Maximum Principle). *Let $\Omega \subset \mathbb{R}^n$ be a domain and let $u \in C^2(\Omega) \cap C(\bar{\Omega})$ be harmonic in Ω . Then, if u achieves its maximum in the interior of Ω it must be constant.*

Proof. Suppose that u achieves its maximum M at an interior point $x_0 \in \Omega$. Define the set

$$X := \{x \in \Omega \mid u(x) = M\} \quad (2.6)$$

Note that by continuity of u this set is closed. Indeed, if $(x_n) \subset \Omega$ is a sequence converging to some $x \in \mathbb{R}^n$ then for all indices n one has easily that $u(x_n) = M$. Thus, we have $\lim_{n \rightarrow \infty} u(x_n) = M = u(x)$ by continuity of u . But then, we have $u(x) = M$ and hence $x \in X$.

Moreover, X is open. This follows from the mean value property. Indeed, if $x \in X$ then we have $u(x) = M$. By the mean value property, we have some small ball $B(x, \rho) \subset \Omega$ so that

$$\oint_{B(x, \rho)} u(y) dy = u(x)$$

which is possible only if $u \equiv M$ in $B(x, \rho)$. In other words, $B(x, \rho) \subset X$ and hence X is open. Consequently, X is clopen and we may then write Ω as the disjoint union of two open sets. One must be empty (because Ω is a domain and thus connected), but by hypothesis we know $X \neq \emptyset$. Thus, $X^c = \emptyset$ and we see that $u \equiv M$ in Ω ; proving our claim. ■

COROLLARY 11. *Under the assumptions of the strong maximum principle.*

$$\max_{\bar{\Omega}} u = \max_{\partial\Omega} u$$

The reader should observe that all of these results are equally valid for holomorphic functions, as $\mathbb{C} \sim \mathbb{R}^2$ and the real and imaginary parts of holomorphisms are harmonic by a previous observation.

THEOREM 12 (Liouville). *If a function $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is harmonic in $\mathbb{R}^n$ and bounded, it is constant.*

Edward Nelson's Proof. We have some constant $M > 0$ so that $|u| \leq M$ for all of $\mathbb{R}^n$. Fix two points $x_1, x_2 \in \mathbb{R}^n$. Suppose without harm that $x_1 \neq x_2$, we will show $u(x_1) = u(x_2)$. Take any balls $B(x_1, R)$ and $B(x_2, R)$ for $R \gg 0$. No matter how large we make the balls, the integrals

$$\oint_{B(x_1, R)} u(y) dy \leq M, \quad \oint_{B(x_2, R)} u(y) dy \leq M \quad (2.7)$$

Now, by making R arbitrarily large we may make symmetric difference $B(x_1, R) \Delta B(x_2, R)$ arbitrarily small and hence make these integrals arbitrarily close. However, these are equal to $u(x_1)$ and $u(x_2)$ respectively. Thus, $u(x_1) = u(x_2)$ and we are done as this pair was arbitrary. ■

2.3 COMPLETENESS OF $\Delta(\Omega)$

The aforementioned argument is applicable in a variety of cases, most notably when there is a “stable” integral characterization of the property at hand. Another important instance is the space of harmonic functions over $\mathbb{R}^d$. We give a result analogous to that of $\mathcal{H}(\Omega)$ below.

THEOREM 13. *Let $\Omega \subseteq \mathbb{R}^d$ be an open set. The space $\Delta(\Omega)$ is complete.*

Proof. Let (f_n) be Cauchy in $\Delta(\Omega)$. Then, for each $n \in \mathbb{N}$ the function f_n is continuous over $\overline{\Omega}$. By the same argument as before, there exists a function $u(x)$ so that $u_n(x) \rightarrow u(x)$ point-wise for all $x \in \Omega$ so that u is the *uniform* limit of u_n with respect to the sup-norm. Yet again, as the uniform limit of a continuous function, we conclude that u is continuous on $\overline{\Omega}$.

We now show that $\Delta u \equiv 0$ in Ω . It is sufficient to establish the mean value integral property. Let $\xi \in \Omega$ be given, and let $B(\xi, \rho)$ be an open ball about ξ contained in Ω . Since $u_n \in \Delta(\Omega)$ for all indices n it is clear that

$$\oint_{B(\xi, \rho)} u_n(x) dx = u_n(\xi), \quad \forall n \in \mathbb{N}$$

By a similar argument, since $\|u_n - u\|_\infty \rightarrow 0$ as $n \rightarrow \infty$ one can establish that

$$\lim_{n \rightarrow \infty} \oint_{B(\xi, \rho)} u_n(x) dx = \oint_{B(\xi, \rho)} u(x) dx \quad (2.8)$$

On the other hand,

$$\lim_{n \rightarrow \infty} \oint_{B(\xi, \rho)} u_n(x) dx = \lim_{n \rightarrow \infty} u_n(\xi) = u(\xi) \quad (2.9)$$

whence,

$$\oint_{B(\xi, \rho)} u(x) dx = u(\xi)$$

■

CHAPTER 3

$\mathcal{H}(\Omega)$ AS A BANACH SPACE

3.1 CONVERGENCE ON RESTRICTIONS OF $\mathcal{H}(\Omega)$

It should be without doubt that $\mathcal{H}(\Omega)$ forms a vector space over $\mathbb{C}$ when endowed with usual addition and scalar multiplication. By Theorem (3) we also have that $\mathcal{H}(\Omega)$ when endowed with the sup norm $\|\cdot\|_\infty := \sup_{\zeta \in \Omega} |\cdot(\zeta)|$ is a Banach space.

There are several *very* surprising facts about $\mathcal{H}(\Omega)$ for an open set $\Omega \subseteq \mathbb{C}$. We have already established that this space is complete in the sup-norm, and therefore $\mathcal{H}(\Omega)$ is closed. It is clear that if $f \in \mathcal{H}(\Omega)$ then $f', f'', \dots, f^{(n)} \in \mathcal{H}(\Omega)$ for all $n \in \mathbb{N}$, since holomorphic functions are infinitely differentiable. What is surprising, is that if $(f_\ell)_\ell$ is a sequence of elements in this space converging to f , then

$$f_\ell^{(n)} \xrightarrow{\mathcal{H}(\Omega)} f^{(n)}, \quad \text{as } \ell \rightarrow \infty \text{ on any compact subset}$$

Before we prove this astounding result, I believe that it is best to prove the following lemma:

LEMMA 14 (Compact Inclusion Lemma). *Let $\Omega \subseteq \mathbb{C}$ be an open set and $\Lambda \subset \Omega$ compact. There exists $\alpha > 0$ such that for all $z \in \Lambda$ the closed ball $\overline{B}(z, \alpha) \subseteq \Omega$.*

Proof. If $\Omega = \mathbb{C}$ it is sufficient to set α equal to 1. Otherwise, Ω^c is non-empty and closed. We argue by contradiction. Instead, there would exist a sequence (z_n) in Λ such that for all indices n $\overline{B}(z_n, 1/n) \cap \Omega^c \neq \emptyset$. By compactness, we may extract a subsequence (z_m) converging to a point $z \in \Lambda$. Now, from each $\overline{B}(z_m, 1/m) \cap \Omega^c$ we select a point $w_m \in \Omega^c$, thereby constructing an associated sequence (w_n) living in Ω^c .

We now claim that $z \in \Omega^c$. It is sufficient to show that $\lim w_n = z$ since Ω^c is closed. To see this, note that for all m :

$$|w_m - z| \leq |w_m - z_m| + |z_m - z| \leq \frac{1}{m} + |z_m - z| \quad (3.1)$$

letting $m \rightarrow \infty$ yields the result, i.e. that $z \in \Omega^c$. But, $z \in \Lambda \subset \Omega$ as well. Contradiction. ■

THEOREM 15. *Let (f_k) be a sequence in $\mathcal{H}(\Omega)$ converging to a function $f \in \mathcal{H}(\Omega)$. Then, the sequence $(f_k^{(n)})$ converges to $f^{(n)}$ on all compact subsets Λ of Ω .*

Proof. It is obviously sufficient to show that $(f'_k) \rightarrow f'$ in the space $\mathcal{H}(\Omega | \Lambda)$, for arbitrary $\Lambda \subset \Omega$ compact. Now, by Lemma (14) it is sufficient to prove that for all $\alpha > 0$ the sequence above converges uniformly upon

$$Z_\alpha := \left\{ z \in \Omega : \overline{B}(z, \alpha) \subseteq \Omega \right\}$$

Let $F : \Omega \rightarrow \mathbb{C}$ be holomorphic. By Cauchy's integral formula, we have immediately that for $z \in \Lambda$

$$\begin{aligned} |F'(z)| &\leq \frac{1}{2\pi} \oint_{\partial B(z, \alpha)} \frac{|fF(\zeta)|}{|\zeta - z|^2} d\zeta = \frac{1}{2\pi} \oint_{\partial B(z, \alpha)} \frac{|F(\zeta)|}{\alpha^2} d\zeta \\ &\leq \frac{2\pi\alpha}{2\pi\alpha^2} \cdot \sup_{\zeta \in \partial B(z, \alpha)} |F(\zeta)| \end{aligned}$$

Hence, we have that $\sup_{z \in \Lambda} |F'(z)| \leq \frac{1}{\alpha} \sup_{\zeta \in \partial B(z, \alpha)} |F(\zeta)|$. Applying this for the function $f_k - f$ we recover

$$\sup_{z \in \Lambda} |f'_k(z) - f'(z)| \leq \frac{1}{\alpha} \sup_{\zeta \in \partial B(z, \alpha)} |f_n(\zeta) - f(\zeta)| \leq \frac{1}{\alpha} \|f_k - f\|_\infty$$

letting $k \rightarrow \infty$ concludes the proof of this theorem. ■

The following lemma is a useful consequence:

LEMMA 16. *Let $\Omega \subseteq \mathbb{C}$ be an open and connected set and suppose $(f_n)_n \subset \mathcal{H}(\Omega)$ is a sequence of injective functions that converge uniformly to f on every compact set $\Lambda \subset \Omega$. If $f : \Omega \rightarrow \mathbb{C}$ is not injective then it must be constant*

Proof. We first observe that since $\mathcal{H}(\Omega)$ is complete it is closed and hence we must have $f \in \mathcal{H}(\Omega)$. Suppose, by way of contradiction, that f is not injective and that f is not-constant; i.e. there exist points $z_1 \neq z_2$ in Ω such that $f(z_1) = f(z_2)$ but f is non-constant. Set $g_n(z) := f_n(z) - f_n(z_1)$ on Ω . Since each f_n is injective, g_n vanishes only at z_1 for all $n \in \mathbb{N}$.

It is easy to see that $g_n \rightarrow g := f - f(z_1)$ on Λ . By hypothesis $g \not\equiv 0$ on Ω . Therefore, by the *Identity Theorem*, z_2 must be an isolated zero: there exists a small circle $\gamma \subset \Omega$ about z_2 such that $g(z)$ vanishes once (at z_2) inside and on γ . By the principle of argument it follows that

$$m = \frac{1}{2\pi i} \oint_{\gamma} \frac{g'(\zeta)}{g(\zeta)} d\zeta$$

where $m \in \mathbb{N}$ is the multiplicity of the zero at z_2 . Now, γ is compact and hence by hypothesis one has that $1/g_n \rightarrow 1/g$ uniformly on γ ; similarly by choosing a small annulus about γ with closure contained entirely in Ω we conclude by Theorem (15) that $g'_n \rightarrow g'$ uniformly on every compact subset of the open annulus, and hence on γ . Furthermore, we may shrink γ such that z_1 is neither inside nor on γ .

Thus,

$$\frac{g'_n}{g_n} \xrightarrow{\mathcal{H}(\Omega|\gamma)} \frac{g'}{g}$$

Now, by uniform convergence by an argument similar to those used in Chapter 1 one can easily see

$$\lim_{n \rightarrow \infty} \frac{1}{2\pi i} \oint_{\gamma} \frac{g'_n(\zeta)}{g_n(\zeta)} d\zeta = \frac{1}{2\pi i} \oint_{\gamma} \frac{g'(\zeta)}{g(\zeta)} d\zeta = m > 0$$

However, g_n has no zeros inside or on γ , whence a final application of the argument principle yields

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{g'_n(\zeta)}{g_n(\zeta)} d\zeta = 0, \quad \forall n \in \mathbb{N}$$

letting $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \frac{1}{2\pi i} \oint_{\gamma} \frac{g'_n(\zeta)}{g_n(\zeta)} d\zeta = 0$$

a contradiction. ■

3.2 THE RIEMANN MAPPING THEOREM

In this section, we shall employ our previous results to prove an extremely powerful theorem. We shall first begin with some definitions, in case the reader is not familiar with the concept of *conformal equivalence*.

Let $U, V \subseteq \mathbb{C}$ be open sets. A function $f : U \rightarrow V$ is called a **conformal mapping**, or **biholomorphism**, if it is both holomorphic in U and bijective. Whenever this holds, the sets U, V are called conformally equivalent. This is an equivalence relation on open sets of $\mathbb{C}$. The cornerstone theorem of this text is the theorem below. Throughout this section we shall write $\mathbb{D}$ to denote the open unit disk about the origin:

$$\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$$

THEOREM 17 (Riemann Mapping Theorem). *Suppose $\Omega \subset \mathbb{C}$ is proper, open and simply connected. If $\xi \in \Omega$ is given, there exists a biholomorphism $\Gamma : \Omega \rightarrow \mathbb{D}$ satisfying*

$$\Gamma(\xi) = 0, \quad \Gamma'(\xi) > 0 \tag{3.2}$$

Moreover, the biholomorphism Γ is uniquely determined.

Some remarks are in order. We note first that we do indeed require that Ω be proper, i.e. $\Omega \neq \mathbb{C}$. Certainly, suppose that we could find a biholomorphism $\Gamma : \mathbb{C} \rightarrow \mathbb{D}$. Then, Γ would be *entire* (holomorphic in all of $\mathbb{C}$) and bounded uniformly there, which would imply that Γ be constant, and hence not injective: a contradiction.

To see that this is true, we give the following analogue to Liouville's theorem seen in the previous chapter

THEOREM 18 (Liouville). *Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be entire and bounded uniformly in $\mathbb{C}$. Then f is constant.*

Proof. Let $z \in \mathbb{C}$, denote the bound of f by $M > 0$ and let $R \gg 0$. Consider the circle $\Gamma_R := \{|\zeta - z| = R\}$. By Cauchy's integral formula we have

$$f'(z) = \frac{1}{2\pi i} \oint_{\Gamma_R} \frac{f(\zeta)}{(\zeta - z)^2} d\zeta, \quad \forall R > 0$$

Therefore, with the parametrization $\zeta(\vartheta) := z + Re^{i\vartheta}$ for $0 \leq \vartheta < 2\pi$:

$$|f'(z)| \leq \frac{1}{2\pi} \int_0^{2\pi} \frac{M}{R^2} \cdot R d\vartheta \sim \mathcal{O}\left(\frac{1}{R}\right) \xrightarrow{R \rightarrow \infty} 0$$

therefore, $f'(z) = 0$ for all $z \in \mathbb{C}$ and hence $f \equiv f(0)$ in $\mathbb{C}$. ■

It follows that the requirement that Ω be proper is justified. Of course, we require simple connectivity of Ω because $\mathbb{D}$ is simply connected, and this is preserved under a biholomorphism.

PROOF OF THE RIEMANN MAPPING THEOREM. We give the proof from [2]. Let $\Omega \subsetneq \mathbb{C}$ be as given in the statement of the theorem. The proof may be subdivided into three crucial observations/steps:

STEP 1. ■

BIBLIOGRAPHY

- [1] E. Stein, R. Shakarchi. *Measure Theory, Integration and Hilbert Spaces*, Princeton Lectures in Analysis, 3rd edition, 2007.
- [2] E. Stein, R. Shakarchi. *Complex Analysis.*, Princeton Lectures in Analysis, 3rd edition, 2007.
- [3] Lars Ahlfors. *Complex Analysis: An Introduction to the Theory of Analytic Functions of One Complex Variable*, 3rd edition 1953.
- [4] Lawrence. C. Evans. *Partial Differential Equations*, AMS Graduate Texts in Analysis, Second Edition.
- [5] E. Chernysh. *A Useful Construction of a Holomorphic Complex Logarithm*, November 2016, <http://cs.mcgill.ca/~echern2/misc/complexLog.pdf>